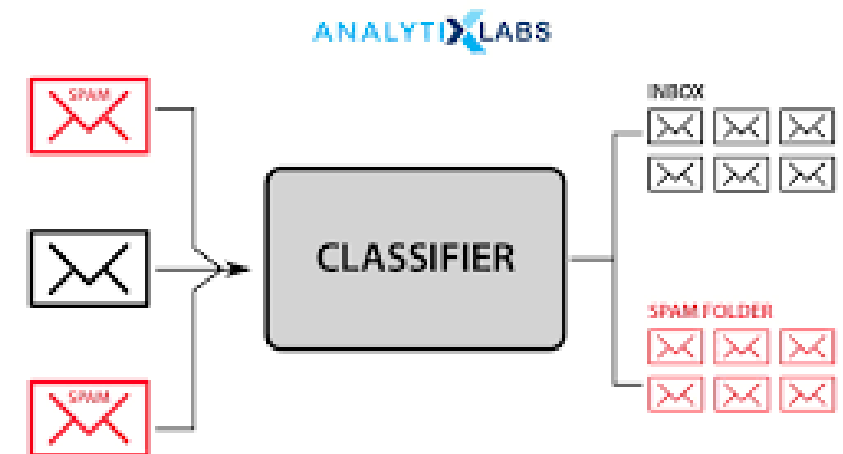
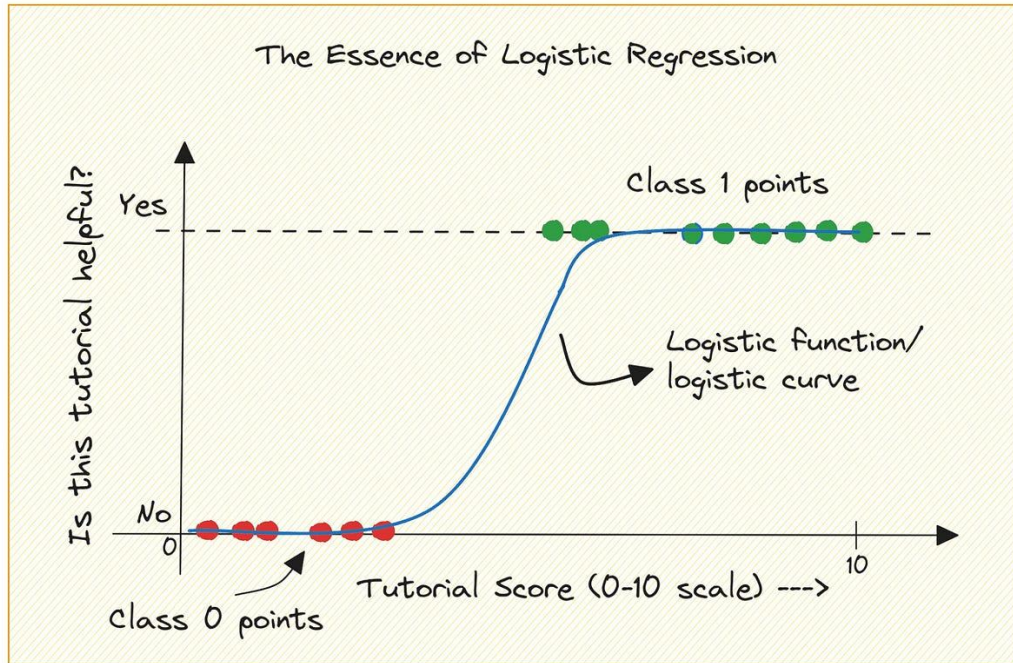

Classification Foundations

Predicting Categories and Making Decisions

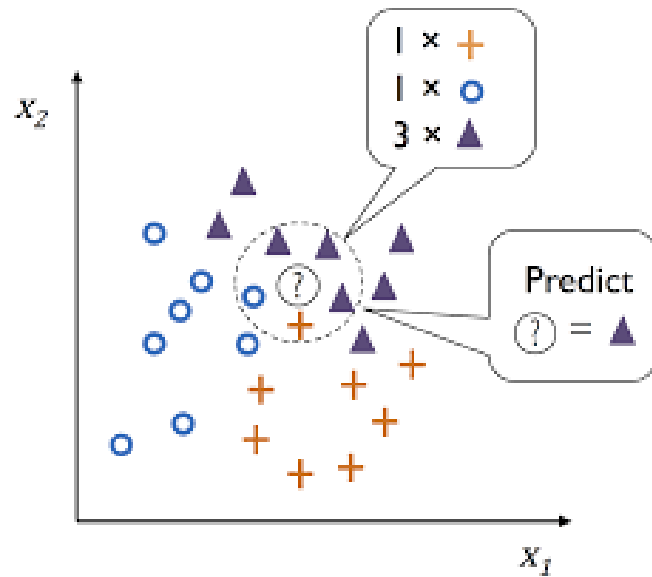
Instructor: Zaryab Ahmad Khan





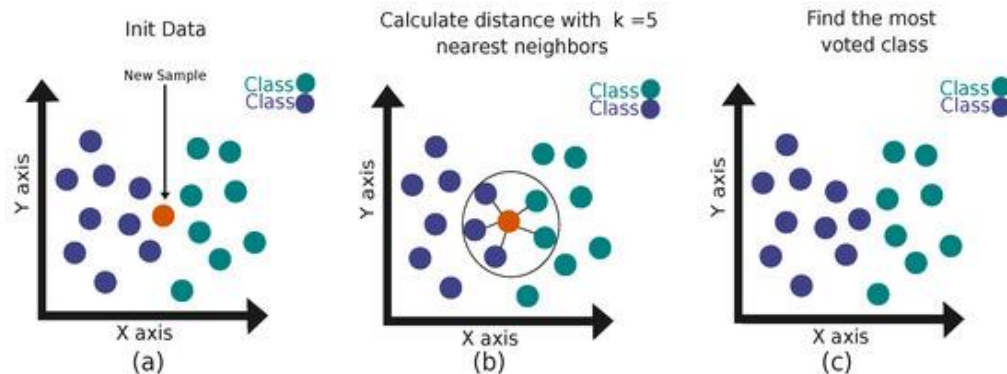
Logistic Regression (The S-Curve)

- **The Concept:** Despite having "Regression" in its name, this is purely for classification. It predicts probabilities between 0 and 1.
- **How it works:** It maps any real number to a value between 0 and 1 using a Sigmoid (S-shaped) curve. If the probability is > 0.5 , it predicts "Yes" (1). Otherwise, "No" (0).
- **Real-World Use:**
 - Is this patient at risk for a specific disease? (Yes/No)
 - Is this transaction fraudulent? (Yes/No)

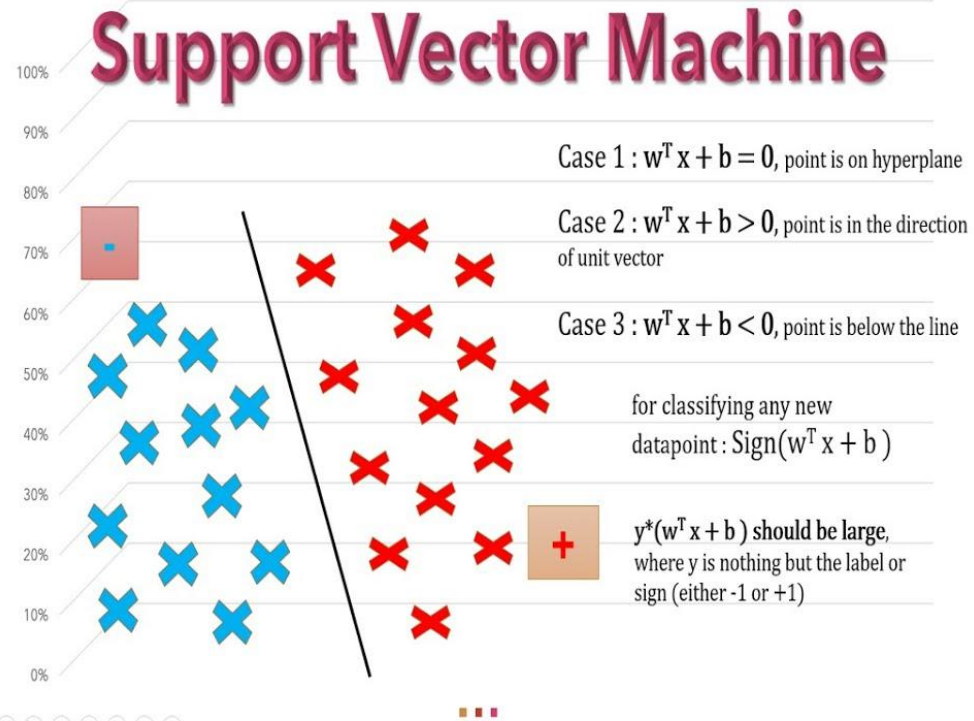


K-Nearest Neighbors (The Democratic Algorithm)

- **The Concept:** "Tell me who your friends are, and I will tell you who you are."
- **How it works:** When given a new data point, the algorithm looks at the K number of data points closest to it on the graph. It takes a "vote" among those neighbors to decide the category.
- **Important:** Always use an odd number for K (like 3, 5, or 7) to avoid tie votes!



Support Vector Machines (The Widest Street)



For positive point :

$$+1 \cdot y * (w^T x + b)$$

For negative point :

$$-1 \cdot y * (w^T x + b)$$

- **The Concept:** Drawing a boundary between categories.
- **How it works:** SVM doesn't just draw a line between two classes; it tries to draw the *widest possible mathematical street* (margin) between them. The data points sitting right on the edge of the street are called the "Support Vectors".
- **Real-World Use:** Highly effective in complex, high-dimensional spaces like image recognition or text categorization.